In this paper, we review in typical spaces in the general topology, from the Language of functions.

「Def.

Given a set X and a family of Topological space $\{Y_i\}_{i \in I}$ with functions

$$\{f_i : X \to Y_i\},$$

the topology $\mathcal{U}$ on X is called the initial topology if

$(X, \mathcal{U})$ is the coarest Topology such that for all $i \in I$, f_i is continuous

Equivalently,

$$\mathcal{U} = \bigcap \{\mathcal{U}' \mid \text{for all } i \in I,\ f_i : (X, \mathcal{U}') \to Y_i \text{ is continuous}\},$$

and generated by the subbasis $\mathcal{S} := \{f_i^{-1}[U] \mid \text{for some } i \in I,\ U \subseteq Y_i \text{ open}\}$.

Dually, a set X and a family of Topological space $\{Y_i\}_{i \in I}$ with functions

$$\{g_i : Y_i \to X\}$$

the topology $\mathcal{U}$ on X is called the final topology if

$(X, \mathcal{U})$ is the finest Topology such that for all $i \in I$, g_i is continuous

Equivalently, the $\mathcal{U}$ is final topology if and only if $\mathcal{U}$ satisfies

$U \subseteq X$ is open set if and only if for each $i \in I$, $g_i^{-1}[U] \subseteq Y_i$ is open.

Moreover, given topological space Z and a map $h : Z \to X$, and $w : X \to Z$,

$h : Z \to (X, \mathcal{U}_{init})$ is continuous if and only if $f_i \circ h : Z \to Y_i$ is continuous, for all $i \in I$

$w : (X, \mathcal{U}_{final}) \to Z$ is continuous if and only if $w \circ g_i : Y_i \to Z$ is continuous for all $i \in I$.

Diagramically,

$$\begin{tikzcd} Z \arrow[r, shift left, "h"] & X \arrow[l, shift left, "w"] \arrow[r, shift left, "f_i"] & Y_i \arrow[l, shift left, "g_i"] \end{tikzcd}$$

Product Space

「Sub space

「Metric space

Let X be a set with a Metric $d: X \times X \to \mathbb{R}$.

The Topology $\mathcal{T}_d$ generated by a basis $\{B_r(x) \mid x \in X, r > 0\}$

is the Coarset topology such that the metric d is continuous.

Prof. Given a basis element $(a,b) \subset \mathbb{R}$, $d^{-1}[(a,b)] = \{(x,y) \in X \times X \mid d(x,y) \in (a,b)\}$.
If $b \leq 0$, then the preimage by d is empty, so open. Now, assume that $b > 0$.
Let $(x,y) \in d^{-1}[(a,b)]$ be given. Then, $a < d(x,y) < b$.
take $r_x, r_y > 0$ such that $r_x + r_y < \min\{d(x,y) - a, b - d(x,y)\}$.
(It is possible, rigorously, take $r_x = r_y = \frac{1}{4} \cdot \min\{d(x,y) - a, b - d(x,y)\}$.)
Then, given $(x', y') \in B_{r_x}(x) \times B_{r_y}(y)$, that is, $d(x, x') < r_x$ and $d(y, y') < r_y$,
By the triangle inequality gives that

$$d(x', y') \leq d(x', x) + d(x, y) + d(y, y') < r_x + d(x, y) + r_y < b$$

and

$$d(x, y) \leq d(x, x') + d(x', y') + d(y', y) < r_x + d(x', y') + r_y$$

$$\Rightarrow a < d(x,y) - (r_x + r_y) < d(x', y'), \text{ so that}$$

$$a < d(x', y') < b, \text{ it means } (x', y') \in d^{-1}[(a,b)].$$

(We also used the symmetricity of metric), therefore

$$(x,y) \in B_{r_x}(x) \times B_{r_y}(y) \subseteq d^{-1}[(a,b)].$$

Since $B_{r_x}(x) \times B_{r_y}(y)$ is open set of $X \times X$, so $d^{-1}[(a,b)]$ is open. ∴ d is continuous.

To show that the second assertion, let $\mathcal{T}'$ be a Topology on X such that
the Metric $d: X \times X \to \mathbb{R}$ is continuous. Let $x_0 \in X$ be fixed.
Since every continuous map is continuous in each variable separately, the map
$f_{x_0}: X \to \mathbb{R};\ x \mapsto d(x, x_0)$ is continuous. We want to show that; $\mathcal{T}_d \subseteq \mathcal{T}'$.
Let $B_r(x) \in \mathcal{T}_d$ be an open set of the metric space. Then, for f_x,

$$B_r(x) = f_x^{-1}[(-\infty, r)]$$

is open set of $\mathcal{T}'$, since f_x is continuous on $\mathcal{T}'$ and $(-\infty, r) \subseteq \mathbb{R}$ is open set. ▨

the initial topology with respect to $\{d_x : X \to \mathbb{R} : y \mapsto d(x,y) \mid x \in X\}$.